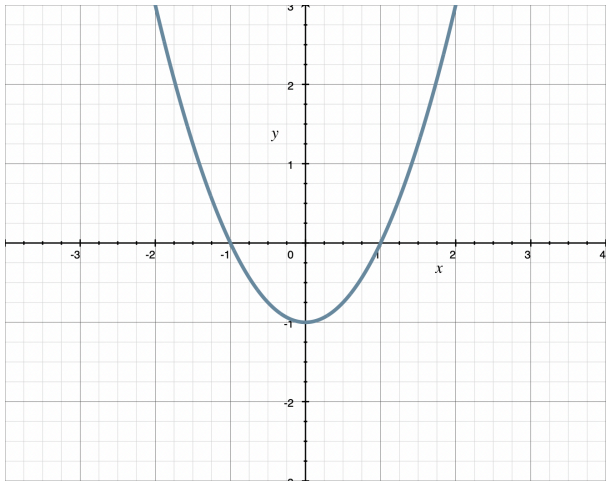
Algebra 2
Final Exam Solutions

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MATH

Algebra 2 Final Exam Answer Key

1. (5 pts) A ■ C D E
2. (5 pts) A ■ C D E
3. (5 pts) A B ■ D E
4. (5 pts) A B C ■ E
5. (5 pts) ■ B C D E
6. (5 pts) A B C D ■
7. (5 pts) A B ■ D E
8. (5 pts) ■ B C D E
9. (15 pts) $-2-i$
10. (15 pts) $(0,-1)$ and $(3,2)$
11. (15 pts)



12. (15 pts) $f(g(x)) = 12x^2 - 68x + 107$

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Algebra 2 Final Exam Solutions

1. B. Calculate the interest using $I = prt$, $p = 200$, $r = 0.02$ (remember that $2\,\% = 0.02$), and $t = 5$.

$$I = (200)(0.02)(5)$$

$$I = 20$$

Now add the interest to the initial deposit of $\$200$.

$$\$200 + \$20$$

There is $\$220$ in the account after 5 years.

2. B. We need to find the common denominator, which will be $3xy^2z^3$. Multiply each term by whatever is required to make the denominator $3xy^2z^3$ (remember you can only multiply by a well chosen 1, which means that whatever you multiply on the bottom you need to also multiply on the top).

$$\frac{y}{3x} + \frac{b}{yz^3} - \frac{c}{3y^2}$$

$$\frac{y}{3x} \cdot \frac{y^2z^3}{y^2z^3} + \frac{b}{yz^3} \cdot \frac{3xy}{3xy} - \frac{c}{3y^2} \cdot \frac{xz^3}{xz^3}$$

$$\frac{y^3z^3}{3xy^2z^3} + \frac{3bxy}{3xy^2z^3} - \frac{cxz^3}{3xy^2z^3}$$

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Now combine the fractions.

$$\frac{y^3z^3 + 3bxy - cxz^3}{3xy^2z^3}$$

3. C. Inverse variation means that the second variable is in the denominator. In other words, the number of red R will equal a constant divided by the number of Blue B.

$$r = \frac{k}{b}$$

Find k by plugging in 4 for r and 1 for b.

$$4 = \frac{k}{1}$$

$$k = 4$$

Find the number of blue when $r = 2$ by plugging 2 in for r and 4 in for k.

$$2 = \frac{4}{b}$$

Solve for b by multiplying both sides by b and dividing by 2.

$$2 \cdot b = \frac{4}{b} \cdot b$$

$$2b = 4$$

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$$\frac{2b}{2} = \frac{4}{2}$$

$$b = 2$$

4. D. Square both sides.

$$\sqrt{x^2 + 5x - 14} = x + 2$$

$$\left(\sqrt{x^2 + 5x - 14}\right)^2 = (x + 2)^2$$

The square and square root will cancel on the left. Use FOIL to expand the right side of the equation.

$$x^2 + 5x - 14 = x^2 + 2x + 2x + 4$$

$$x^2 + 5x - 14 = x^2 + 4x + 4$$

Solve for x.

$$x^2 - x^2 + 5x - 14 = x^2 - x^2 + 4x + 4$$

$$5x - 14 = 4x + 4$$

$$5x - 4x - 14 = 4x - 4x + 4$$

$$x - 14 = 4$$

$$x - 14 + 14 = 4 + 14$$

$$x = 18$$

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5. A. Let's label the equations to make things more clear.

[1] $2x + 3y - z = 17$

[2] $3x - y + 2z = 11$

[3] $x - 3y + 3z = -4$

Start by eliminating z from equation [2] by multiplying [1] by 2.

$2(2x + 3y - z = 17)$

$4x + 6y - 2z = 34$

Then add this to [2].

$4x + 6y - 2z + (3x - y + 2z) = 34 + 11$

$7x + 5y = 45$

Next, eliminate z from equation [3] by multiplying [1] by 3.

$3(2x + 3y - z = 17)$

$6x + 9y - 3z = 51$

Then add this to [3].

$6x + 9y - 3z + (x - 3y + 3z) = 51 + (-4)$

$7x + 6y = 47$

Subtract $7x + 5y = 45$ from $7x + 6y = 47$ to eliminate x.

$7x + 6y - (7x + 5y) = 47 - 45$

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$$7x - 7x + 6y - 5y = 47 - 45$$

$$y = 2$$

Find x by plugging 2 in for y into the equation $7x + 5y = 45$.

$$7x + 5(2) = 45$$

$$7x + 10 = 45$$

$$7x + 10 - 10 = 45 - 10$$

$$7x = 35$$

$$\frac{7x}{7} = \frac{35}{7}$$

$$x = 5$$

Find z by plugging 5 in for x and 2 in for y into the equation $2x + 3y - z = 17$.

$$2(5) + 3(2) - z = 17$$

$$10 + 6 - z = 17$$

$$16 - z = 17$$

$$16 - 16 - z = 17 - 16$$

$$-z = 1$$

$$z = -1$$

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6. E. Remember that the standard form of a quadratic expression is $ax^2 + bx + c$. For the equation $6x^2 - 11x + 4$: $a = 6$, $b = -11$, and $c = 4$. Multiply $a \cdot c$ and find factors of the result that combine to b.

$$6 \cdot 4 = 24$$

We'll make a table with all of the factors, and their sum.

Factors of 24	Sum
-1, -24	-25
-2, -12	-14
-3, -8	-11
-4, -6	-10

$-3 + -8 = -11$, so they're the factors we're looking for. Now we'll divide each factor by a and reduce if possible.

$$\frac{-3}{6} = \frac{-1}{2}$$

One factor of the quadratic is $(2x - 1)$ because the denominator of the reduced fraction becomes the coefficient to x, and then add or subtract the numerator depending on the sign (in this case we'll subtract since -1 was the numerator).

$$\frac{-8}{6} = \frac{-4}{3}$$

The other factor of the quadratic is $(3x - 4)$ because the denominator of the reduced fraction becomes the coefficient to x,

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and then add or subtract the numerator depending on the sign (in this case we'll subtract since -4 was the numerator).

$$(2x-1)(3x-4)$$

7. C. Switch x and y in the original equation.

$$y=\sqrt{x-2}+3$$

$$x=\sqrt{y-2}+3$$

Solve for y.

$$x-3=\sqrt{y-2}+3-3$$

$$x-3=\sqrt{y-2}$$

$$(x-3)^2=\sqrt{y-2}^2$$

$$x^2-3x-3x+9=y-2$$

$$x^2-6x+9=y-2$$

$$y-2+2=x^2-6x+9+2$$

$$y=x^2-6x+11$$

8. A. Combine the logarithms using the quotient rule.

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$$\log_4 \frac{32}{2}$$

$$\log_4 16$$

Convert from the form $y = \log_b x$ to $b^y = x$.

$$4^y = 16$$

$$y = 2$$

9. Simplify the powers of i by remembering that $i^2 = -1$.

$$\frac{4 - 3i}{1i^2 + 2i^5}$$

$$\frac{4 - 3i}{1(-1) + 2(-1)(-1)i}$$

$$\frac{4 - 3i}{-1 + 2i}$$

Use the conjugate method to get the imaginary number out of the denominator.

$$\frac{4 - 3i}{-1 + 2i} \cdot \frac{-1 - 2i}{-1 - 2i}$$

$$\frac{(4 - 3i)(-1 - 2i)}{(-1 + 2i)(-1 - 2i)}$$

Use the FOIL method to multiply the binomials in the numerator and denominator.

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$$\frac{-4-8i+3i+6i^2}{1+2i-2i-4i^2}$$

$$\frac{-4-5i+6i^2}{1-4i^2}$$

Plug -1 in for i^2.

$$\frac{-4-5i+6(-1)}{1-4(-1)}$$

$$\frac{-4-6-5i}{1+4}$$

$$\frac{-10-5i}{5}$$

$$-2-i$$

10. Use the second equation and solve for y.

$$y-x=-1$$

$$y-x+x=-1+x$$

$$y=x-1$$

Plug $x-1$ in for y into the first equation and solve for x.

$$(y-2)^2+x^2=9$$

$$(x-1-2)^2+x^2=9$$

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$$(x-3)^2+x^2=9$$

$$x^2-6x+9+x^2=9$$

$$2x^2-6x+9=9$$

$$2x^2-6x+9-9=9-9$$

$$2x^2-6x=0$$

$$\frac{2x^2}{2}-\frac{6x}{2}=\frac{0}{2}$$

$$x^2-3x=0$$

$$x(x-3)=0$$

$x=0$ and $x=3$

Plug 0 in for x into the equation where y has already been isolated.

$$y=0-1$$

$$y=-1$$

Plug 3 in for x into the equation where y has already been isolated.

$$y=3-1$$

$$y=2$$

The solutions are:

$(0,-1)$ and $(3,2)$

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11. Draw the x- and y-axes and make sure to label tick marks and axes. The graph of $f(x) = x^2 - 1$ will be a parabola that opens up. The y-intercept can be found by substituting $x = 0$ in the equation and solving for y.

$$y = (0)^2 - 1 = -1$$

So put a point on the y-axis at -1. To find the x-intercept(s), set $y = 0$ and solve for x.

$$0 = x^2 - 1$$

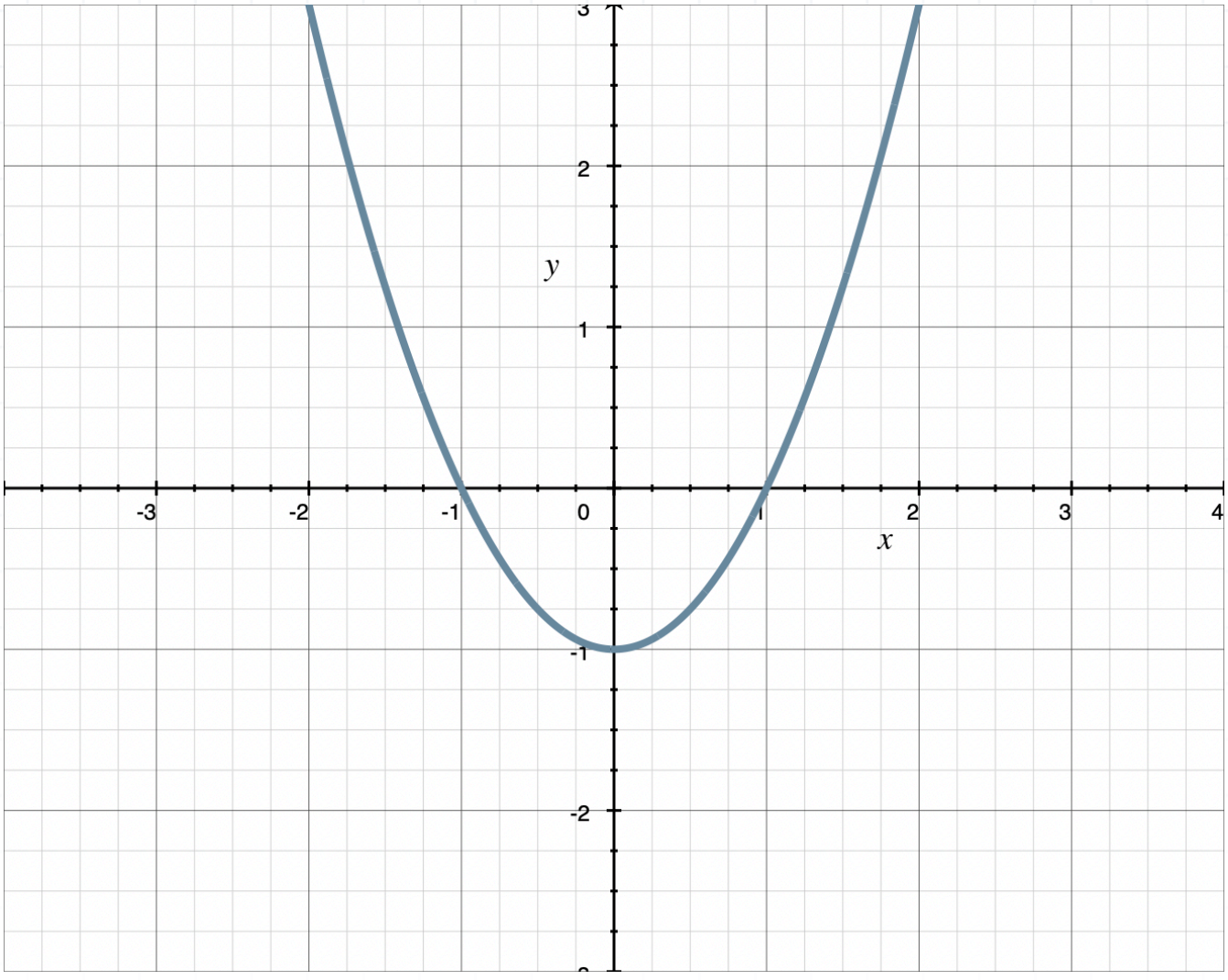
$$1 = x^2$$

$$\pm\sqrt{1} = \sqrt{x^2}$$

$$\pm 1 = x$$

Plot points on x-axis at -1 and 1. Then we can sketch the parabola.

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12. Plug $2x - 5$ in for x into the $f(x)$ equation.

$$f(g(x)) = 3(2x - 5)^2 - 4(2x - 5) + 12$$

Simplify.

$$f(g(x)) = 3(4x^2 - 20x + 25) - 8x + 20 + 12$$

$$f(g(x)) = 12x^2 - 60x + 75 - 8x + 32$$

$$f(g(x)) = 12x^2 - 68x + 107$$

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